

IRELE

Advanced Measurement and Data Driven Modelling

Lab report

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Contents

Lab 1: RLS with forgetting factor

1.1 Main steps

1. Measure the input and output signals from the DUT. The input is arbitrary and the time varying part of the system is due to a manual switch.
2. Make a first estimate of the parameters with an LLS estimator
3. Use the RLS estimator with forgetting factor to track the time-varying parameters of the system.
4. Deduce from the estimated parameters the time instants at which the system changes.

1.2 Time-varying RLS estimation

Starting from the recursive equation of the system, a matrix formulation is derived:

$$y(N) = b_0(N)u(N) + b_1(N)u(N-1) - a_1(N)y(N-1) - a_2(N)y(N-2) \quad (1.1)$$

$$= K(N)\theta(N) \quad (1.2)$$

with

$$K(N) = \begin{bmatrix} u(N) & u(N-1) & -y(N-1) & -y(N-2) \end{bmatrix}, \quad \theta(N) = \begin{bmatrix} b_0(N) \\ b_1(N) \\ a_1(N) \\ a_2(N) \end{bmatrix} \quad (1.3)$$

The RLS estimator (with forgetting factor) is a recursive algorithm that computes the parameters $\theta(N)$ and their covariance matrix K_N after each new sample is collected. Because it is recursive, it needs to be initialized and this is done with a linear least squares estimate on a few first samples.

The linear least squares estimator, used to find θ in $y_{\text{lls}} = H\theta$, is in our case constructed as follows:

$$H = \begin{bmatrix} u(n_{\text{start}} + 1) & u(n_{\text{start}}) & -y(n_{\text{start}}) & -y(n_{\text{start}} - 1) \\ u(n_{\text{start}} + 2) & u(n_{\text{start}} + 1) & -y(n_{\text{start}} + 1) & -y(n_{\text{start}}) \\ \vdots & \vdots & \vdots & \vdots \\ u(n_{\text{LLS}} - 1) & u(n_{\text{LLS}} - 2) & -y(n_{\text{LLS}} - 2) & -y(n_{\text{LLS}} - 3) \end{bmatrix}, \quad y_{\text{lls}} = \begin{bmatrix} y(n_{\text{start}} + 1) \\ y(n_{\text{start}} + 2) \\ \vdots \\ y(n_{\text{LLS}} - 1) \end{bmatrix} \quad (1.4)$$

Where $n_{\text{start}} = \max(n_a, n_b)$ and n_{LLS} is the number of samples used for the LLS estimate, chosen to be 200 here.

$\theta(n_{\text{LLS}})$ and $P_{n_{\text{LLS}}}$ are found with, respectively, the solution to the LLS problem and the definition of the covariance matrix in the RLS algorithm:

$$\theta(n_{\text{LLS}}) = (H^T H)^{-1} H^T y_{\text{lls}}, \quad P_{n_{\text{LLS}}} = (H^T H)^{-1} \quad (1.5)$$

The Matlab code of the LLS initialization is given below:

```
% initialization with an LLS estimate over a few of the first samples
% y_lls = H * theta
H = zeros(sampleForLLS - max(na, nb), na + nb + 1);
for uCol = 0:nb
    H(:, uCol + 1) = u((max(na, nb) + 1 - uCol:sampleForLLS - uCol));
end
for yCol = 1:na
    H(:, nb + 1 + yCol) = -y((max(na, nb) + 1 - yCol:sampleForLLS - yCol));
end
y_lls = y((max(na, nb) + 1:sampleForLLS));

theta = H \ y_lls;
P = inv(H' * H);
```

The recursive least squares equations with forgetting factor g are given by:

$$\theta(N) = \theta(N-1) - \frac{P_{N-1} K^T(N) [K(N) \theta(N-1) - y(N)]}{g + K(N) P_{N-1} K^T(N)} \quad (1.6)$$

$$P_N = \frac{1}{g} \left(P_{N-1} - \frac{P_{N-1} K^T(N) K(N) P_{N-1}}{g + K(N) P_{N-1} K^T(N)} \right) \quad (1.7)$$

The Matlab code implementing these equations is given below:

```
theta_sav = zeros(na + nb + 1, N - max(na, nb) + 2);
P_diag_sav = zeros(na + nb + 1, N - max(na, nb) + 2);

for itr = sampleForLLS+1:N
```

```

% saving previous values
theta_sav(:, itr - max(na, nb) + 1) = theta;
P_diag_sav(:, itr - max(na, nb) + 1) = diag(P);

K_itr = [u(itr - (0:nb)), -y(itr - (1:na))];

theta = theta - (P * (K_itr')) * (K_itr * theta - y(itr)) / (g + K_itr * P * (K_itr'));
P = 1/g * (P - (P * (K_itr')) * K_itr * P) / (g + K_itr * P * (K_itr'));

end

```

1.3 Results

The input of the system is not interesting to show here as it is random white noise. The evolution in time of the estimated parameters is however much more interesting. The following figures will display the evolution of the parameters (with their uncertainty range) for multiple forgetting factor values and for different experiments. The experiments are :

1. Excitation of a LTI sub-system (namely Low LTI) - Figure ??.
2. Excitation of the second LTI sub-system (namely High LTI) - Figure ??.
3. Excitation of both LTI systems trough a single switch - Figure ??.
4. Excitation of both LTI systems trough multiple switches - Figure ??.

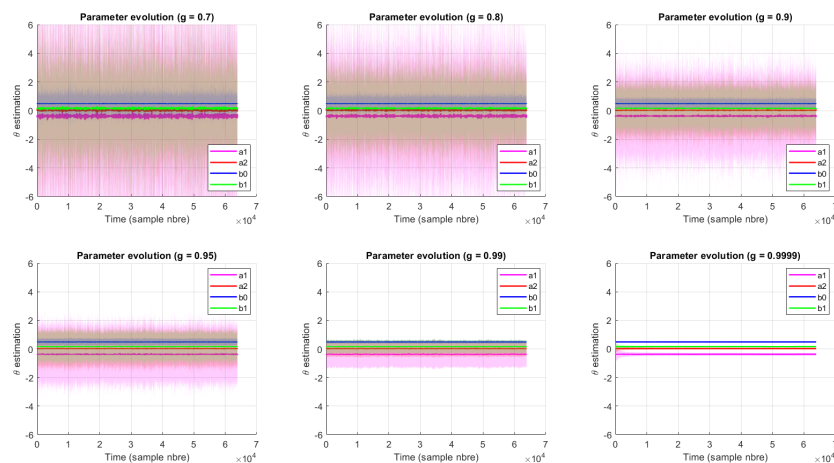


Figure 1.1: Parameters evolution for multiple forgetting factors - Low LTI sub-system

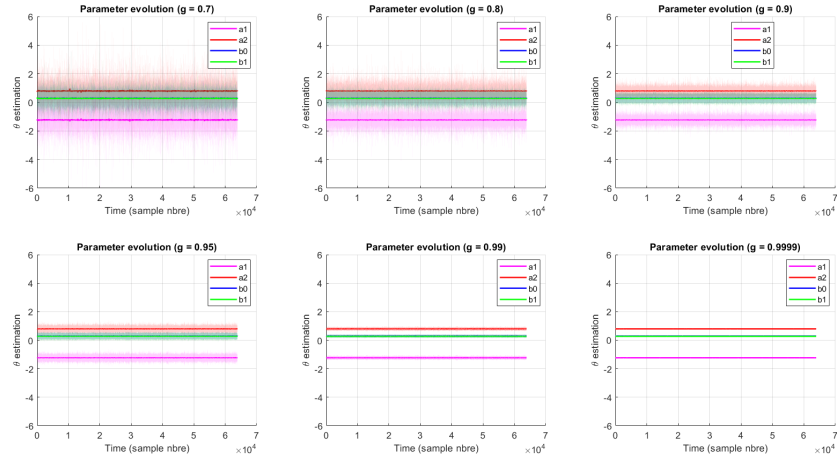


Figure 1.2: Parameters evolution for multiple forgetting factors - High LTI sub-system

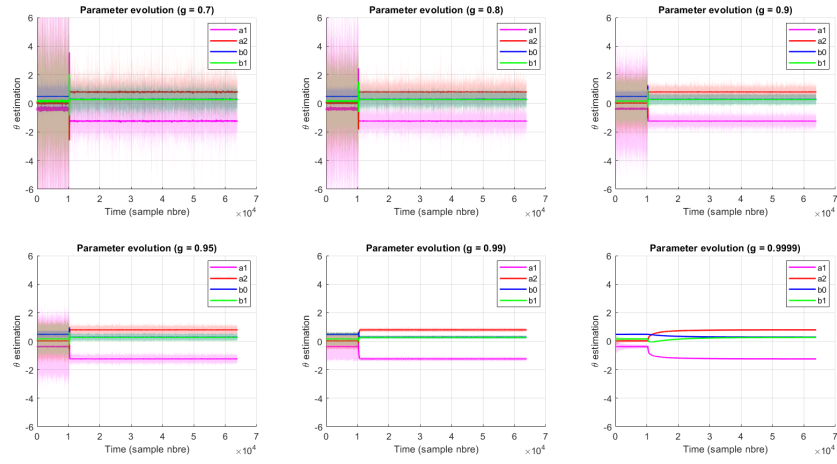


Figure 1.3: Parameters evolution for multiple forgetting factors - single switch between the two LTI systems

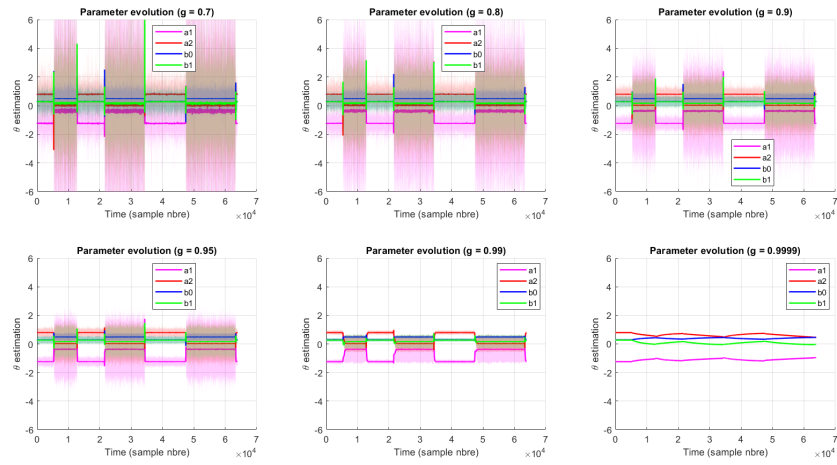


Figure 1.4: Parameters evolution for multiple forgetting factors - multiple switches between the two LTI systems

One may already observe that the uncertainty on the parameters of the LTI system named High LTI system is around 4 times lower than the uncertainty on the parameters for the second LTI system.

Another observation is related to the reduction of the uncertainty converging to a plateau value as the time passes at the beginning of the measurement. This behaviour is well visible in the case of the Low LTI subsystem for a forgetting factor set at 0.9999. The reason behind the uncertainty level diminution is laying in the limited number of samples available for the RLS algorithm at the beginning of the measurements - Figure ??.

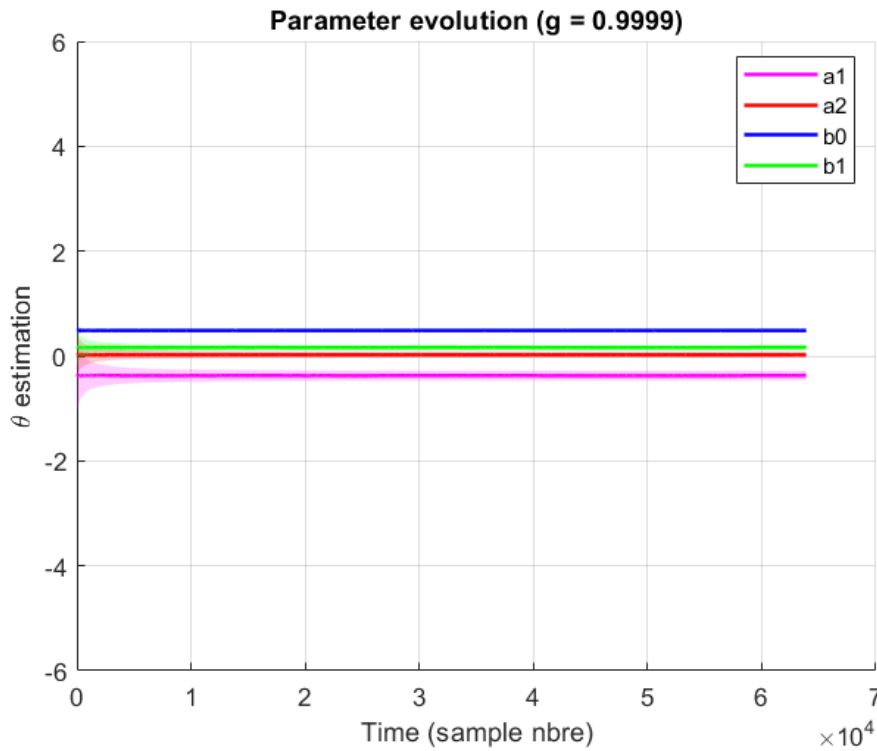


Figure 1.5: Uncertainty variation during the first 500 samples - Low LTI system ($g=0.9999$)

As shown on the figures, the RLS algorithm with forgetting factor is a powerful tool to track the parameters of time-varying system simulated in our case by switching two LTI subsystems. Nevertheless, the choice of the forgetting factor value is dependent of the constraints or specifications of our system and also the required confidence on the parameters we get once in steady-state.

Let's consider our experiment with multiple switches - One can easily see that the RLS algorithm with a forgetting factor of 0.9999 cannot converge fast enough and is therefore not relevant for such time-varying system.

In the other hand, the other RLS algorithm with forgetting factor smaller than 0.9999 exhibits higher parameter variances leading to uncertainty much more significant (10 times higher in case of $g = 0.99$); but also noticeable spikes at the switching instants resulting to a non-smooth evolution of the parameters. This rough evolution may result to potential issues, damages in real systems we are tracking (in case by example

of RLS algorithm used for control system). For rigorousness, a table containing the parameters variance for the different forgetting factors is shown below - Table ?? and ??.

Low LTI	σ_{a1}^2	σ_{a2}^2	σ_{b0}^2	σ_{b1}^2
g=0.7	46.0729	19.1852	0.3077	13.5596
g=0.8	19.5318	8.1543	0.1653	5.5176
g=0.9	7.2197	2.7688	0.0767	1.9273
g=0.95	3.5704	1.2898	0.0393	0.9333
g=0.99	0.2875	0.1140	0.0090	0.0791
g=0.9999	$0.2385e^{-3}$	$0.2123e^{-3}$	$0.1125e^{-3}$	$0.1429e^{-3}$

Table 1.1: Parameters variances in steady state for Low LTI system

High LTI	σ_{a1}^2	σ_{a2}^2	σ_{b0}^2	σ_{b1}^2
g=0.7	1.2416	1.4443	2.0137	2.7133
g=0.8	0.5458	0.6222	0.7531	0.9635
g=0.9	0.1918	0.1995	0.2014	0.2356
g=0.95	0.0874	0.0842	0.0694	0.0798
g=0.99	0.0204	0.0189	0.0117	0.0135
g=0.9999	$0.4765e^{-3}$	$0.3578e^{-3}$	$0.1062e^{-3}$	$0.1921e^{-3}$

Table 1.2: Parameters variances in steady state for High LTI system

It remains therefore to decide the most suitable forgetting factor for our experiment with multiple switches. In our case, the forgetting factor of 0.99 was converging few hundreds of samples after the switch (0.15 seconds) to its steady-state - fast enough to converge before the next switching and with a smooth convergence. The switching instants and the convergence of the parameters to its steady-sate is shown in Figure ??.

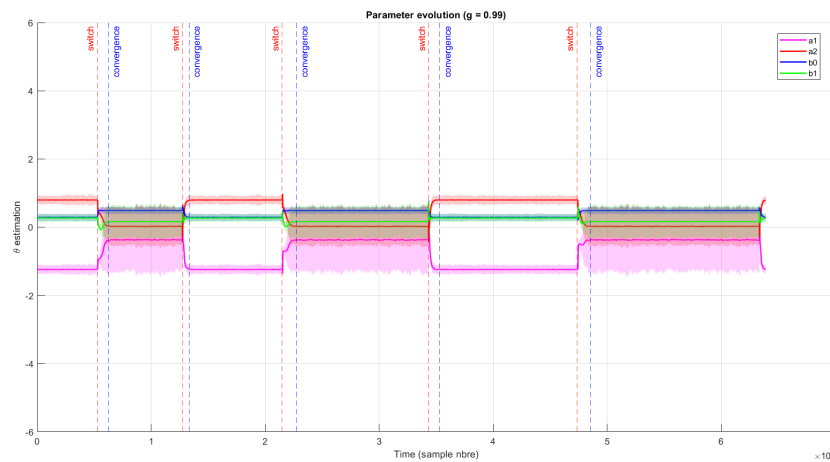


Figure 1.6: Switching instants and convergence (g=0.99)

In order to grasp the differences between the results given for different forgetting factor, it is useful to plot the transfer function modeled from the parameters given by the RLS algorithms and compare it with an estimate of the FRF computed using spectral averaging.

We can easily see that RLS algorithm with forgetting factor smaller than 0.99 have reached the convergence

and match the estimated FRF (which is logical as those algorithm are reacting faster than the algorithm with $g=0.99$). Nevertheless, by looking carefully to the modeled transfer functions coming from RLS with forgetting factors smaller than 0.99, one can observe the models are fitting the FRF with a lower accuracy resulting from the higher uncertainty level on the parameters.

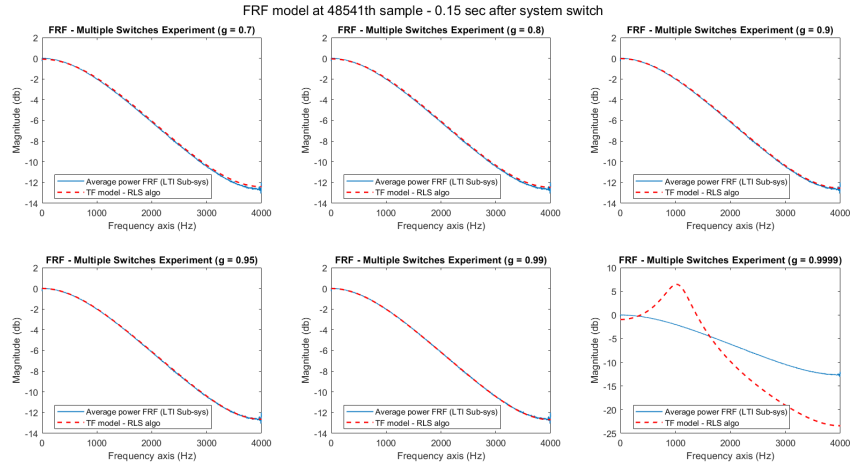


Figure 1.7: Comparison of the estimated FRF and the RLS based estimate after 150 ms

Finally, the comparison between the estimated FRF for both LTI sub-systems and the transfer function calculated based on the parameters - computed by the RLS algorithm with forgetting factor of 0.99 once the steady-state is reached - are displayed in Figure ?? and ??

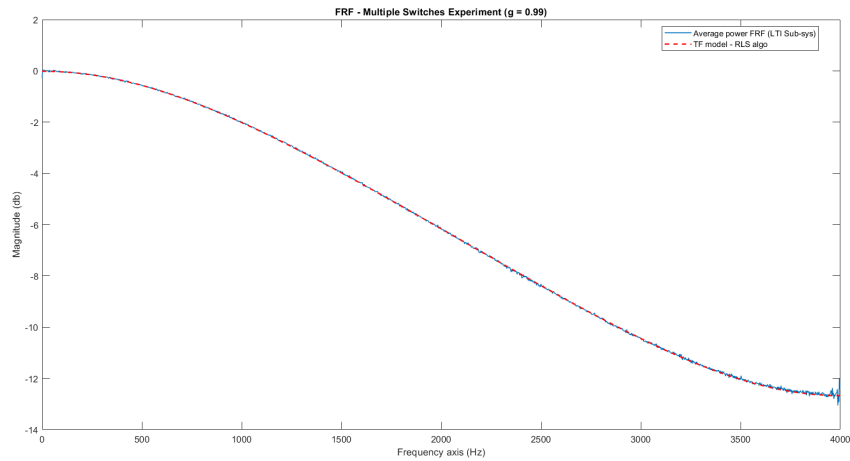


Figure 1.8: Comparison estimated FRF and computed model - Low LTI system - $g=0.99$

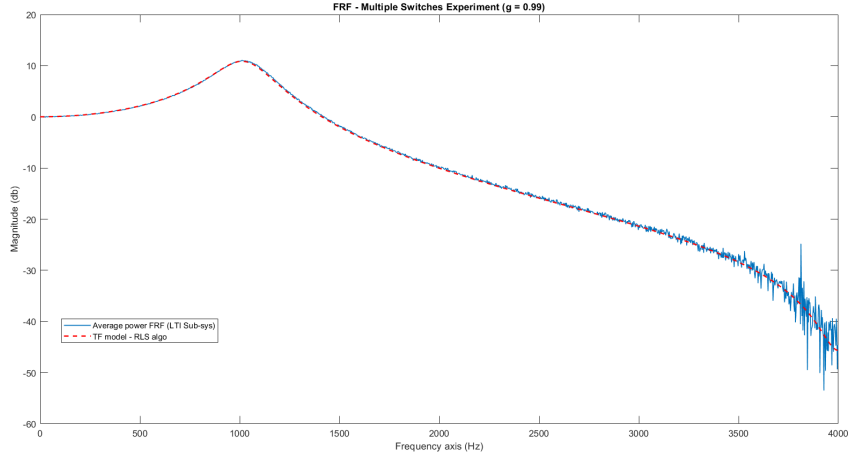


Figure 1.9: Comparison estimated FRF and computed model - High LTI system - $g=0.99$

Those parametric estimates (RLS - $g=0.99$ - steady-state instants) are detailed below:

$$G_{low}(z^{-1}) = \frac{b_0 + b_1 z^{-1}}{1 + a_1 z^{-1} + a_2 z^{-2}} = \frac{0.4874 + 0.1613 z^{-1}}{1 - 0.3763 z^{-1} + 0.0263 z^{-2}} \quad (1.8)$$

$$G_{high}(z^{-1}) = \frac{b_0 + b_1 z^{-1}}{1 + a_1 z^{-1} + a_2 z^{-2}} = \frac{0.2871 + 0.2713 z^{-1}}{1 - 1.2378 z^{-1} + 0.7951 z^{-2}} \quad (1.9)$$

In comparison the transfer function computed from LLS for each LTI sub-system taken separately :

$$G_{low-LLS}(z^{-1}) = \frac{b_0 + b_1 z^{-1}}{1 + a_1 z^{-1} + a_2 z^{-2}} = \frac{0.4832 + 0.1752 z^{-1}}{1 - 0.3518 z^{-1} + 0.01 z^{-2}} \quad (1.10)$$

$$G_{high-LLS}(z^{-1}) = \frac{b_0 + b_1 z^{-1}}{1 + a_1 z^{-1} + a_2 z^{-2}} = \frac{0.2884 + 0.2659 z^{-1}}{1 - 1.2419 z^{-1} + 0.7964 z^{-2}} \quad (1.11)$$

1.4 Conclusion

The RLS algorithm with forgetting factor is able to converge towards the true parameters of a time-varying system, under the condition that the forgetting factor is well chosen. This choice was easy with this system as it was known that it was switching between two LTI systems. In a more complex case, knowing whether the system is varying slowly or the forgetting factor is too high may be difficult.

Lab 2: Kalman filter

2.1 Main steps

1. Obtain the matrices A_c , B_c , C_c and D_c of the state-space model representing the dynamics of the RLC circuit above, in function of the parameters R_0 , L_0 and C_0 . (You can start by applying Kirchoff's circuit laws in the Laplace domain). Write down the steps performed in the report.
2. Show the continuous- and discrete-time step responses, and motivate your choice for the sample time. Write down the discretized matrices A_d , B_d and C_d in the report. Is the system stable? And is it observable?
3. Simulate the system, using the sampling time T_s selected above and selecting the simulation time such that your measurement will have (around) 200 points. Choose the input $u(t)$ as white Gaussian noise with standard deviation 1 V. Set (in the time-delay block) the initial condition $x_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. Add the names of the blocks in the Simulink and add a screenshot of it in your report. Extract the true state of the system from the Simulink model.
4. Compute the state estimates using method a (prediction only) and method b (Kalman filter) - 5 variations.
5. Comments the results in your report, making sure to provide an answer to the following questions. Which method performs better? Which method is more robust? Is there a difference in performance between the estimate of the first state (voltage capacitor) and of the second state (current) for method a? And for method b? Why? What are the effects of the covariance matrices on the Kalman gain?
6. Draw some conclusions on what we have observed, highlighting the advantages and limitations of the Kalman filter that you have seen and comparing it to the other state estimation methods we have analyzed. Add one or two suggestions for practical situations in which you would use the Kalman filter.

2.2 Continuous-time state space model

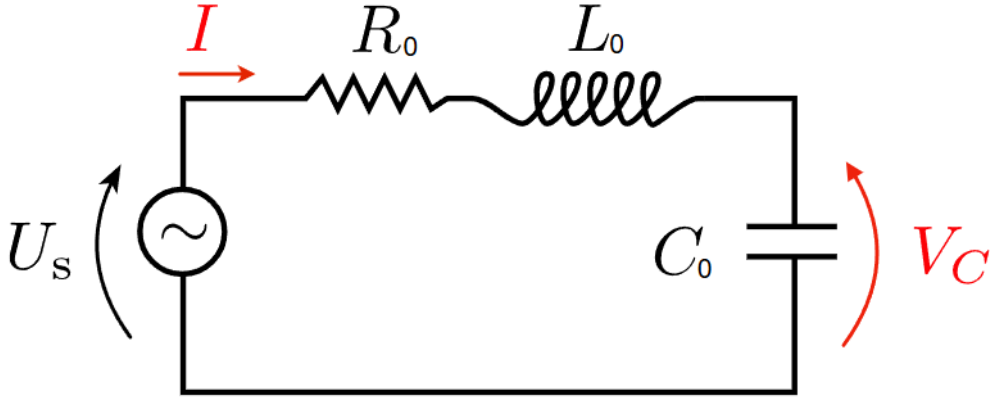


Figure 2.1: Studied RLC circuit

The studied system is the RLC circuit shown in Fig ???. A set of equations can be derived using Kirchhoff's laws:

$$u_s(t) = R_0 i(t) + L_0 \frac{di(t)}{dt} + \frac{1}{C_0} \int i(t) dt \quad (2.1)$$

$$v_C(t) = \frac{1}{C_0} \int i(t) dt \quad (2.2)$$

By substituting the second equation into the first one, we obtain:

$$u_s(t) = R_0 i(t) + L_0 \frac{di(t)}{dt} + v_C(t) \quad (2.3)$$

To get to the continuous-time state space representation of this circuit, we need to define the input of the system as $u_s(t)$, its output as $v_C(t)$ and the state vector as:

$$x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} v_C(t) \\ i(t) \end{bmatrix} \quad (2.4)$$

And using:

$$\begin{cases} \frac{dv_C(t)}{dt} = \frac{i(t)}{C_0} \\ \frac{di(t)}{dt} = -\frac{R_0}{L_0} i(t) - \frac{v_C(t)}{L_0} + \frac{u_s(t)}{L_0} \end{cases} \quad (2.5)$$

Which finally leads to:

$$\frac{d}{dt} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{C_0} \\ -\frac{1}{L_0} & -\frac{R_0}{L_0} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{L_0} \end{bmatrix} u_s(t) \quad (2.6)$$

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} \quad (2.7)$$

$$\Leftrightarrow \quad A_c = \begin{bmatrix} 0 & \frac{1}{C_0} \\ -\frac{1}{L_0} & -\frac{R_0}{L_0} \end{bmatrix}, \quad B_c = \begin{bmatrix} 0 \\ \frac{1}{L_0} \end{bmatrix}, \quad C_c = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D_c = 0 \quad (2.8)$$

2.3 Discrete-time state space model

To convert this continuous-time state space model to a discrete-time one, we need to select a sampling frequency. As the poles of the system are the eigenvalues of the matrix A_c , we can compute them with Matlab, giving:

$$\begin{cases} \lambda_1 \approx (-1 + j8.1035) \times 10^6 \\ \lambda_2 \approx (-1 - j8.1035) \times 10^6 \end{cases} \quad (2.9)$$

The resonance frequency can be deduced from the eigenvalues computed above.

$$\omega_r = 8.1035 \times 10^6 \text{ rad/s} \quad (2.10)$$

$$= 1.29 \text{ MHz} \quad (2.11)$$

The band of interest is therefore chosen to be higher than 1.29 MHz in order to capture the dynamics of the system. Considering our system being a second-order low pass system, an appropriate sampling frequency would be 10 times higher than ω_r . Thus f_s is set to $f_s = 12.9$ MHz. The discrete-time state space matrices are computed with Matlab. Two different methods are used: ZOH and tustin.

To compare which method is closer to the continuous-time model, the step response of the three models are plotted in Fig ??.

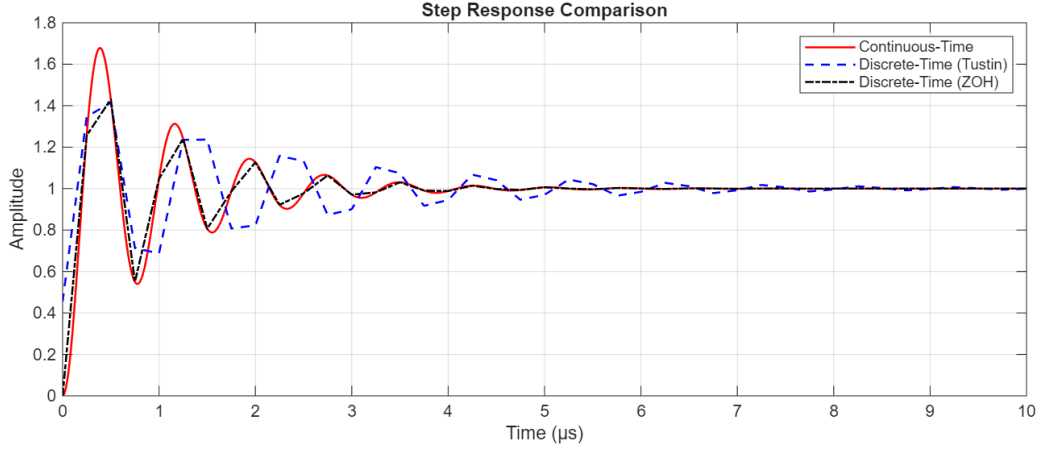


Figure 2.2: Step response comparison between continuous-time model and discrete-time models obtained with ZOH and Tustin methods - 4 MHz

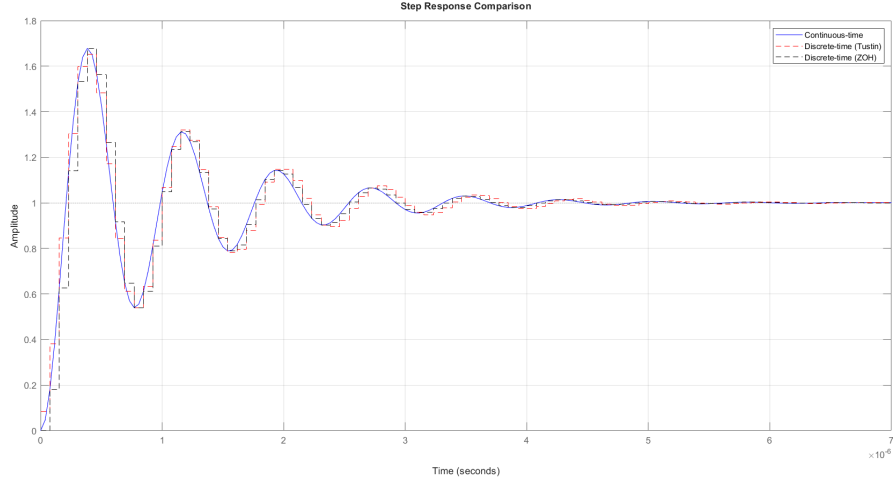


Figure 2.3: Step response comparison between continuous-time model and discrete-time models obtained with ZOH and Tustin methods - 12.9 MHz

The step response with $f_s = 4$ MHz shows that tustin model presents a mismatch with the continuous-time model (delay introduction in time domain), which is not the case for the ZOH model. The ZOH model is therefore chosen for the following steps. The discrete-time state space matrices are:

$$A_d = \begin{bmatrix} 0.8184 & 22.2414 \\ -0.0133 & 0.6849 \end{bmatrix}, \quad B_d = \begin{bmatrix} 0.1816 \\ 0.0133 \end{bmatrix}, \quad C_d = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D_d = 0 \quad (2.12)$$

The poles of the discrete-time system are also computed by calculating the eigenvalues of the matrix A_d and the system is stable as they are in the unit circle:

$$\begin{cases} p_1 \approx -0.7517 + 0.5407i \\ p_2 \approx -0.7517 - 0.5407i \end{cases} \quad (2.13)$$

To determine whether the system is observable, we can compute the rank of the observability matrix $\mathcal{O}$:

$$\mathcal{O} = \begin{bmatrix} C_d \\ C_d A_d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0.8184 & 22.2414 \end{bmatrix} \quad (2.14)$$

Which is of rank 2, meaning that the full space is observable and therefore the states can be fully reconstructed from the output measurement. Kalman filter can therefore reconstructs both states from the output.

2.4 Simulink simulation

The Simulink model is shown in Fig ???. The input is white Gaussian noise with zero mean and variance of 1V. There is also process noise $R_{v_{sim}}$ and measurement noise $R_{n_{sim}}$ which are respectively equal to:

$$R_{v_{sim}} = \begin{bmatrix} 0.025 & 0 \\ 0 & 10^{-6} \end{bmatrix}, \quad R_{n_{sim}} = 0.0025 \quad (2.15)$$

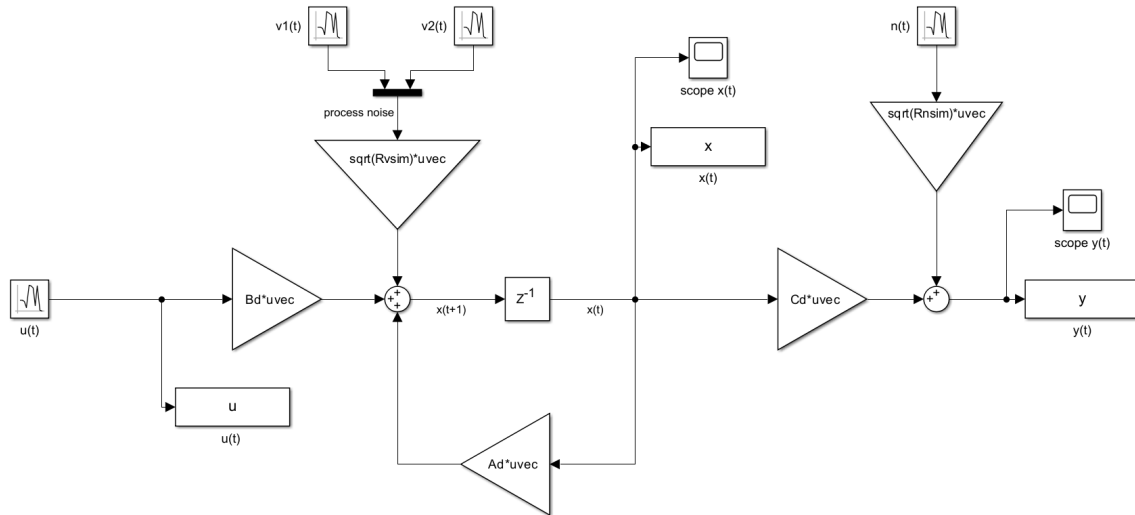


Figure 2.4: Simulink model of the discrete-time state space system

2.5 State estimation

The state estimation is performed in two ways: a simple prediction based on the model, and a Kalman filter. Both methods are implemented in Matlab as follows:

```
for itr = 1:size(xin, 1)-1
    % prediction only
    xPred(:, itr+1) = A * xPred(:, itr) + B * uin(itr, :);

    % Kalman
    Q = A * P(:, :, itr) * A' + Rv; % prediction covariance
    K(:, itr+1) = Q * C' / (C * Q * C' + Rn); % Kalman gain
```

```

P(:, :, itr+1) = (eye(size(A, 1)) - K(:, itr+1) * C) * Q; % updated covariance

prediction = A * xKalman(:, itr) + B * uin(itr, :);
measurement = (yin(itr+1) - C*prediction - D*uin(itr+1, :));

xKalman(:, itr+1) = prediction + K(:, itr+1) * measurement;
end

```

And those state estimates are compared to the true state, obtained from the Simulink model. The results are shown in Fig ?? . It can be observed that the error on the state estimation V_c is approximately 25 dB lower when using the Kalman filter compared to the simple prediction method (open loop observer). In the other hand, the difference between the normalized error for the second state I with Kalman prediction and the open loop observer is much less - around 10dB. The reason will be explained below in the discussion section.

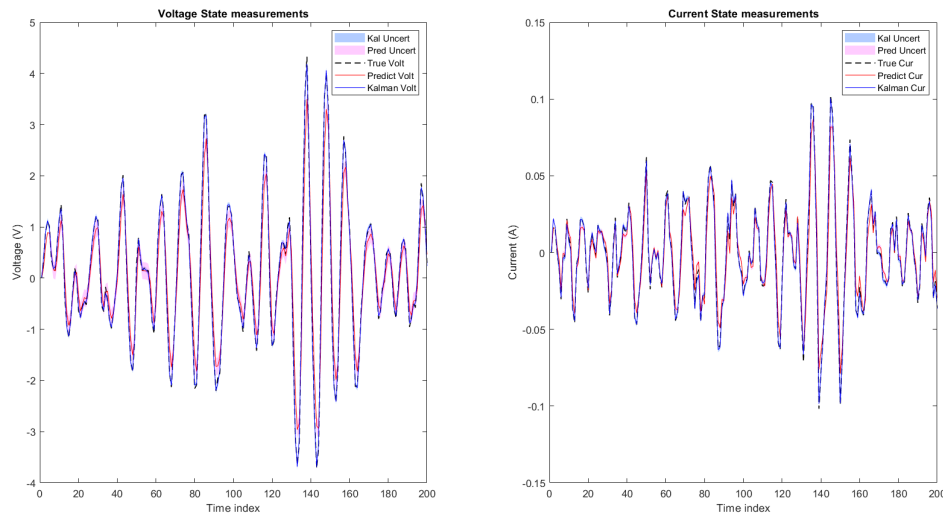


Figure 2.5: State estimation comparison between simple prediction and Kalman filter

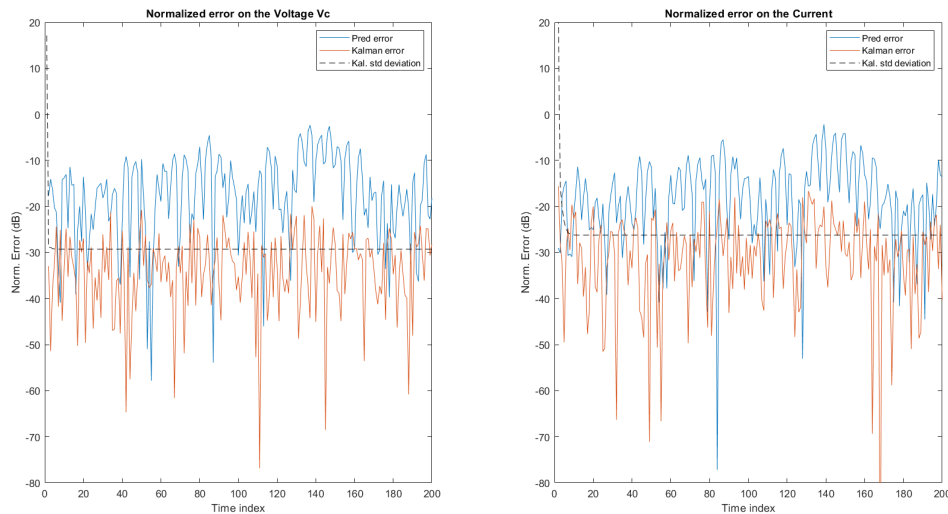


Figure 2.6: Normalized error comparison between simple prediction and Kalman filter

To compare the robustness of both methods, the true parameters of the system are modified and the state estimation is performed again. Only the normalized state error plots are shown.

On fig ?? the introduced error on the model ($A = 2A_{\text{true}}$) makes the model unstable - $|\text{Re}(\lambda_{\text{obs}})| > 0$. The Kalman filter allows to maintain a stable state thanks to its feedback mechanism (the gain K moves the eigenvalues in the Z-domain and make the observer stable).

The impact of a wrong initial condition x_0 is shown in Fig ??. Both methods seem to have no significant difference compared to the case with exact initial conditions on fig ??.

Fig ?? and Fig ?? show the effect of wrong process and measurement noise covariances. In both cases, the estimates are not significantly affected, but the estimated standard deviation is higher than with the true parameters. The standard deviation increase will be also discussed in the conclusion section.

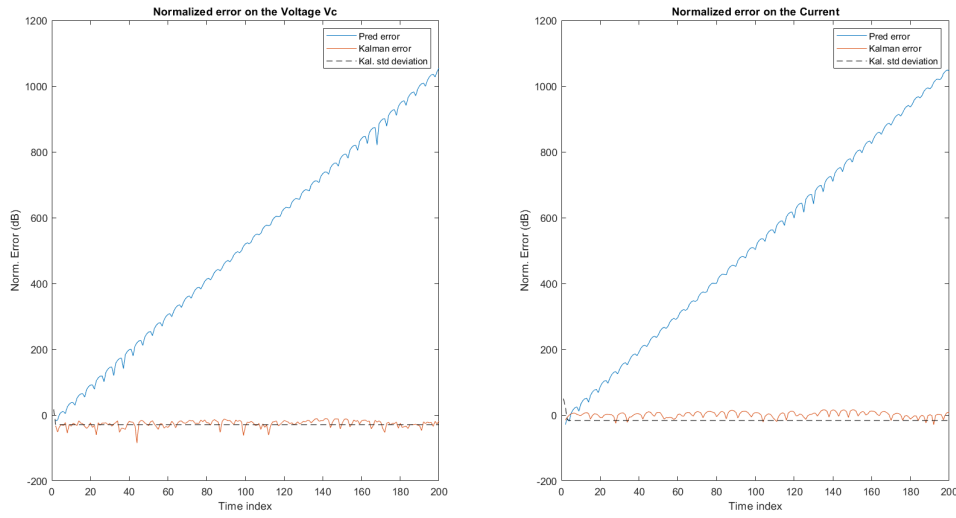


Figure 2.7: State estimation comparison with $A = 2A_{\text{true}}$

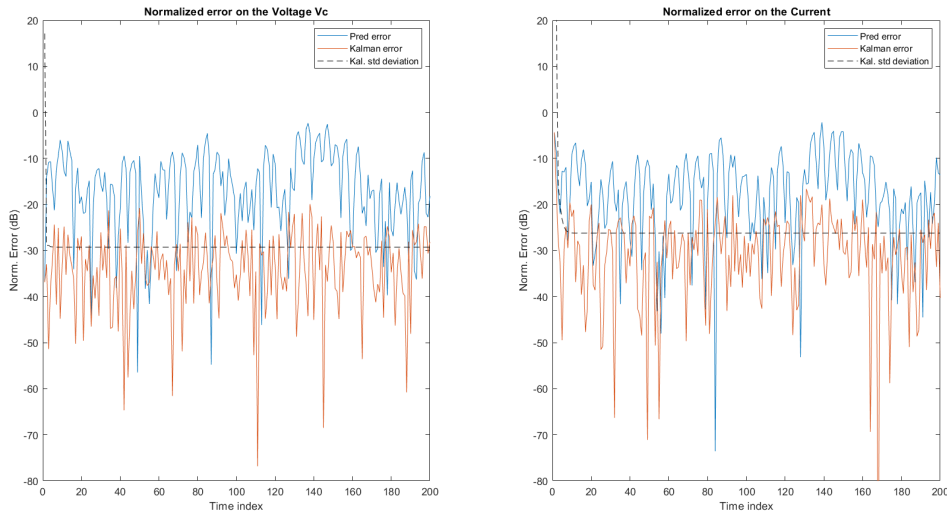


Figure 2.8: State estimation comparison with $x_0 = [0.02; 0.02]$ instead of $x_0 = [0; 0]$

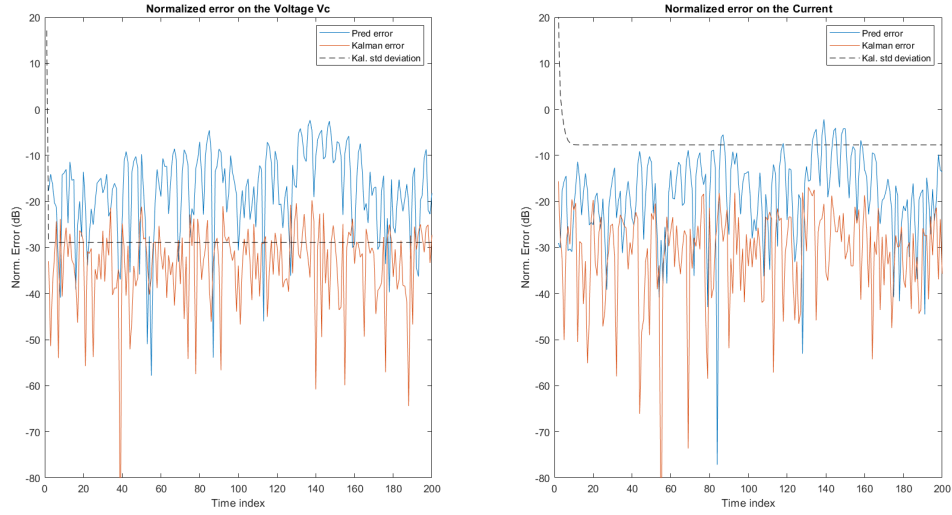


Figure 2.9: State estimation comparison with $R_v = 100R_{v_{\text{true}}}$

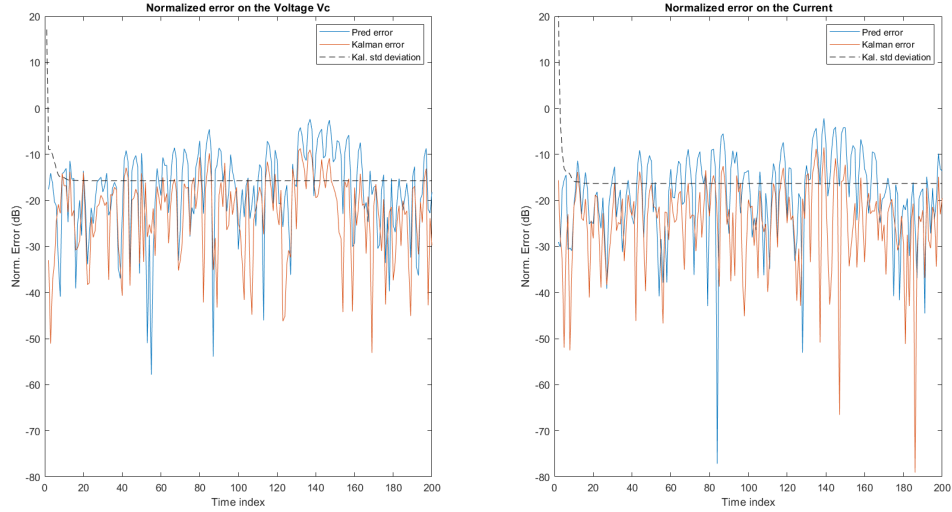


Figure 2.10: State estimation comparison with $R_n = 100R_{n_{\text{true}}}$

2.6 Conclusions

2.6.1 Which method performs better?

Method b (the Kalman filter) performs better. In the experiments run from matlab the Kalman filter produces much smaller state estimation error and converges to the true state independent of the initial condition, whereas the open-loop prediction remains biased and has larger steady-state error. (In our test runs the Kalman filter reduced the estimation error by a margin on the order of tens of dB relative to the open-loop prediction.)

2.6.2 Which method is more robust?

The Kalman filter is more robust in the presence of measurement noise and uncertain initial states: it adapts its estimates using incoming measurements and reduces bias that the predictor cannot correct. However, the Kalman filter is sensitive to model mismatch and to incorrect tuning of the noise covariances: large model errors degrade its performance (but typically it still outperforms the naive predictor).

In short, Kalman = more robust to noise and initial state, but still requires a reasonably accurate model and sensible covariance tuning.

2.6.3 Is there a difference in performance between the estimate of the first state (capacitor voltage) and of the second state (current) for method a? And for method b? Why?

Method a. Naive prediction - open loop observer

The capacitor voltage and current exhibit comparable state estimation accuracy, both showing similar normalized error levels. One might expect the current state estimate to show a higher sensibility to noise than the voltage state estimate (as the capacitor voltage is calculated through current integration and would tend to low pass high frequency components as introduced by the noise) . Nevertheless, the discrete-time state model matrix A_d shows a strong coupling between the states:

$$A_d = \begin{bmatrix} 0.8184 & 22.2414 \\ -0.0133 & 0.6849 \end{bmatrix} \quad (2.16)$$

In particular, the voltage state at $t+1$ exhibit a high dependence with the previous current state at time t compared to the previous voltage state (the off-diagonal parameter $a_{12} = 22.2414$ is much higher than $a_{11} = 0.8184$!). Because of this strong coupling, the error in current state estimation is propagated to the voltage state one step later leading to similar error levels in both estimations.

Method b. Kalman prediction

As explained previously, the Kalman predictor behaves as a closed-loop observer. Unlike the open-loop predictor it continuously corrects its state estimates using the information contained in the output measurement. Since the discrete-time state output matrix C_d is

$$C_d = \begin{bmatrix} 1 & 0 \end{bmatrix} \quad (2.17)$$

Only the capacitor voltage V_c is directly measured at the output. The current state estimation is corrected indirectly, in a same pattern as done with the open loop, through the dynamic coupling with the voltage state. This dynamic coupling exhibiting a weak coupling as shown by the off-diagonal component $a_{21} = -0.0133$:

$$A_d = \begin{bmatrix} 0.8184 & 22.2414 \\ -0.0133 & 0.6849 \end{bmatrix} \quad (2.18)$$

The error on the current state estimate is therefore reduced compared to the naive prediction but to a lesser extent compared to the voltage state which benefit from the direct measurement feedback.

2.6.4 What are the effects of the covariance matrices on the Kalman gain?

The Kalman gain K depends on the covariance of the predicted state Q , the measurement matrix C and the measurement noise covariance R_n via following relation

$$K_{t+1} = Q_{t+1} C^\top (C Q_{t+1} C^\top + R_n)^{-1}.$$

and where covariance on the estimated state P_{t+1} and covariance on the predicted state Q_{t+1} are computed using following relationships:

$$P_{t+1} = (I - K_{t+1} C) Q_{t+1}$$

$$Q_{t+1} = A P_t A^\top + R_v.$$

Thus, intuitively:

- Increasing the measurement noise R_n makes the denominator larger and therefore reduces K : the filter trusts the model more and the noisy measurements less.
- Increasing the covariance on the predicted state Q tends to increase K : larger uncertainty in the state estimate makes the filter rely more on measurements.
- Increasing the process noise covariance R_v increases Q and hence increases K .
- Initial P affects the transient gains.

Edge cases: as $R_n \rightarrow 0$ the gain tends to the (pseudo-)inverse, while as $R_n \rightarrow \infty$ the gain tends to zero and the filter behaves like a simple predictor.

2.6.5 Advantages, limitations and suggestions for practical situations in which you would use the Kalman filter ?

The feedback mechanism of the Kalman filter offers multiple advantages.

Accuracy improvement, stabilization and automation As seen previously, the Kalman prediction owns the benefit of the closed loop observer which allow to place the eigenvalues in the unitary circle thanks to its closed-loop correction mechanism.

This feedback modifies the dynamics of the observer through the Kalman gain K , effectively shifting the observer poles through:

$$A_{obs} = A - KC \quad (2.19)$$

With K computed based on the estimated uncertainty level encoded in the state covariance matrix P .

It is a significant advantage compared to open loop observer which stability depends directly on the poles of the matrix A_d (leading to instability in the case of $A_d = 2xA_d$) but also to other closed loop observers like the Luenberger observers where the poles placement are moved manually by the designer's / user's choice - $(A + LC)$.

Those properties make the Kalman observer automatically stable with moderate noise but also highly flexible to noise variation.

Robustness to the initial state errors and noise The previous simulations shows also a robustness to incorrect initial state conditions. Choosing a sufficiently large initial covariance P at the time instant t_0 impose a high uncertainty on the initial state to the observer. Kalman observer will therefore emphasize the measurement output to estimate the state at the time instant $t + 1$.

The modulation of the gain K through the covariance matrix P (and indirectly Q) secures a quick convergence to the true value even with wrong x_0 .

In addition, the observer will adjust the gain K in function of the process noise covariance and the measurement noise covariance giving an optimal balance between the model and the measurement information. Nevertheless, this specific link between R_v and R_n may lead to some limitations..

Limitations Although it is a powerful tool to track the state, Kalman filter has also limitations :

- We just discussed about how robust is the Kalman observer to the noise but this property is true if we assume that R_v and R_n reflect the reality. If the noise covariances matrices are purely tuned, the observer will return biased estimates or may become unstable. Also the $R_n = 100xR_n$ simulation highlights how excessive noises levels drag the observer to behave like an open-loop observer.
- The second limitation is inherent to any observers. The observation requires observability. It means that states are only recoverable if they are observable from the output y . Additionally, an observer relies on the correctness of the state space representation (A_d, B_d, C_d, D_d) . Kalman predictor cannot compensate for complete unfaithful representation to the system reality even if it shows robustness for limited mismatch ($A_d = 2xA_d$ experiment).

Practical situations

Today, Kalman predictor is used in many situations. We discussed during the course about Artemis program and the rocket trajectory correction but Kalman is also used today for much simpler experiments :

- Automating watering fields with limited number of moisture sensors.
- Robotic control system (Hidden state estimation).
- Autonomous driving (localization).
- Finance (Inflation rates calculation through measured income data by example)
- etc.